## Matrices as data

A grayscale image can be represented by $A \in \mathbb{R}^{m \times n}$, with pixel intensities as entries. Color images usually form tensors; a matrix treatment must specify how the channels are arranged.

Rank-one matrix:

$$uv^T = \begin{bmatrix} u_1 \\ \vdots \\ u_m \end{bmatrix} \begin{bmatrix} v_1 & \cdots & v_n \end{bmatrix},$$

$$(uv^T)_{ij} = u_i v_j.$$

A rank-one factorization stores $m + n$ scalars rather than $mn$ entries. SVD expresses a general matrix as a weighted sum of such pieces. Large singular values often capture substantial reconstruction energy.



## Statement of the SVD

Every

$$A \in \mathbb{R}^{m \times n}, \qquad \operatorname{rank} A = r,$$

admits a full SVD

$$A = U \Sigma V^T.$$

$$U \in \mathbb{R}^{m \times m}, \qquad V \in \mathbb{R}^{n \times n},$$

where $U$ and $V$ have orthonormal columns. The nonzero diagonal entries of the $m \times n$ matrix $\Sigma$ are

$$\sigma_1 \geq \cdots \geq \sigma_r > 0.$$

Reduced SVD:

$$A = U_r \Sigma_r V_r^T,$$

$$U_r \in \mathbb{R}^{m \times r}, \quad \Sigma_r \in \mathbb{R}^{r \times r}, \quad V_r \in \mathbb{R}^{n \times r}.$$

The remaining entries of the full rectangular $\Sigma$ are zero. The reduced SVD retains only the $r$ positive singular values and their vectors.

## Right singular vectors from $A^TA$

$$x^TA^TAx = \|Ax\|^2 \geq 0.$$

Thus $A^TA$ is symmetric positive semidefinite and has spectral decomposition

$$A^TA = V\Lambda V^T.$$

Order the positive eigenvalues and set

$$\lambda_1 \geq \cdots \geq \lambda_r > 0,$$

$$\sigma_i = \sqrt{\lambda_i}.$$

For $i \leq r$,

$$\|Av_i\|^2 = v_i^TA^TAv_i = \sigma_i^2,$$

so $Av_i \neq 0$.

The vectors $v_1, \ldots, v_r$ form an orthonormal row-space basis. The vectors $v_{r+1}, \ldots, v_n$ form an orthonormal nullspace basis.



## Left singular vectors

Define

$$u_i = \frac{Av_i}{\sigma_i}.$$

For $i \neq j$, orthogonality follows from

$$\sigma_i\sigma_ju_i^Tu_j = v_i^TA^TAv_j = \lambda_jv_i^Tv_j = 0.$$

The lengths satisfy

$$\|u_i\|^2 = \frac{\|Av_i\|^2}{\sigma_i^2} = 1.$$

Consequently,

$$Av_i = \sigma_iu_i.$$

The vectors $u_1, \ldots, u_r$ form an orthonormal column-space basis. Complete them with $u_{r+1}, \ldots, u_m$, an orthonormal basis of the left nullspace.

To complete the existence proof, let $V = [v_1, \ldots, v_n]$ be an orthonormal eigenbasis of $A^TA$. For $i > r$,

## Left singular vectors (continued)

$$\|Av_i\|^2 = v_i^T A^T A v_i = 0$$

so $Av_i = 0$. Extend $u_1, \ldots, u_r$ to a complete orthonormal basis of $\mathbb{R}^m$. For any $x = \sum_{j=1}^{n} c_j v_j$,

$$Ax = \sum_{j=1}^{r} c_j A v_j = \sum_{j=1}^{r} c_j \sigma_j u_j = U \Sigma V^T x.$$

This verifies the right vectors, left vectors, singular values, and factorization together.

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https://github.com/xinhi6716/computer-science-self-study



## Comparing $A^T A$ and $AA^T$

$$AA^T = U \Sigma \Sigma^T U^T.$$

The two Gram matrices have the same positive eigenvalues and rank. The multiplicity of zero can differ because their sizes can differ.

This explains the general structure.

- The eigenvectors of $A^T A$ give $V$.
- The eigenvectors of $AA^T$ give $U$.
- The positive eigenvalues are the squared singular values.
- Rectangular and rank-deficient matrices also admit SVD.

Numerical implementations usually avoid forming $A^T A$, which squares the relevant condition number and can overflow. Stable methods use bidiagonal reduction and algorithms for the resulting bidiagonal singular-value problem.

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## Rank-one expansion

Reduced SVD:

$$A = \sum_{i=1}^{r} \sigma_i u_i v_i^T.$$

The Frobenius inner product is $\langle B, C \rangle_F = \operatorname{tr}(B^T C) = \sum_{ij} B_{ij} C_{ij}$, and $\|B\|_F = \sqrt{\sum_{ij} B_{ij}^2}$. Distinct rank-one terms are orthogonal in this inner product.

$$\langle u_i v_i^T, u_j v_j^T \rangle_F = 0, \quad i \neq j.$$

Consequently,

$$\|A\|_F^2 = \sum_i \sigma_i^2.$$

Retaining the first $k$ terms preserves the energy fraction

$$\frac{\sum_{i=1}^{k} \sigma_i^2}{\sum_{i=1}^{r} \sigma_i^2}.$$



## Rank-one expansion (continued)

This ratio requires $A \neq 0$. The nuclear norm is $\|A\|_* = \sum_i \sigma_i$, which differs from the Frobenius norm. Reconstruction energy need not match perceptual relevance, so an image application must specify its error metric and task.

## Truncated SVD: retained directions and discarded errors

Let a real matrix A have rank r and decreasing positive singular values. Restrict approximations to rank at most k, initially with $1 \leq k < r$. Truncation retains the leading terms:

$$A_k = \sum_{i=1}^{k} \sigma_i u_i v_i^T, \qquad E = A - A_k = \sum_{i=k+1}^{r} \sigma_i u_i v_i^T.$$

Retained directions satisfy $A_k v_i = A v_i$. Discarded directions satisfy $A_k v_i = 0$, so $E v_i = \sigma_i u_i$. This constructs a candidate; it has not yet ruled out every other rank-k matrix.

Fix $A = \operatorname{diag}(9, 4, 1)$, k=1. Then $A_1 = \operatorname{diag}(9, 0, 0)$ and $E = \operatorname{diag}(0, 4, 1)$. First distinguish the two error measures, then prove their lower bounds. Here $u_i = v_i = e_i$; a general matrix can have different left and right directions.



## Two norms ask different questions about the same error

For $E = \operatorname{diag}(0, 4, 1)$:

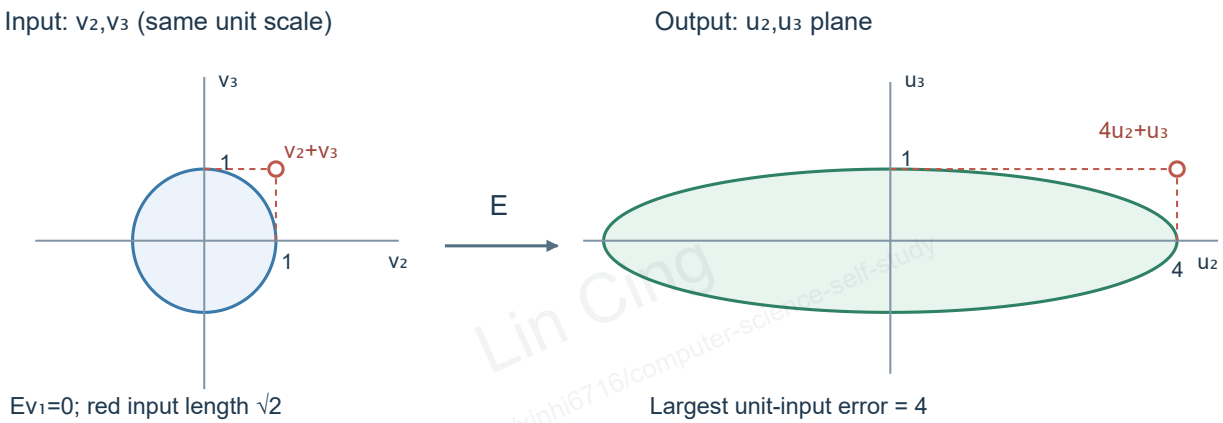
| Unit input | Error output | Squared length |
|---|---|---|
| $e_1$ | 0 | 0 |
| $e_2$ | $4e_2$ | 16 |
| $e_3$ | $e_3$ | 1 |

The spectral norm asks for the **largest error over all unit inputs**. If $\|x\| = 1$,

$$\|Ex\|^2 = 16x_2^2 + x_3^2 \leq 16(x_1^2 + x_2^2 + x_3^2) = 16.$$

Input $e_2$ attains equality, hence $\|E\|_2 = 4$. The Frobenius norm instead sums squared column errors: $\|E\|_F^2 = 0 + 16 + 1 = 17$, so $\|E\|_F = \sqrt{17}$. It is not another maximum unit-input output length.

## Error ellipse: why $\sqrt{17}$ is not its longest semiaxis




Since $Ev_1 = 0$, project the three-dimensional unit ball onto the v2,v3 unit disk, then apply E. Its image is the ellipse disk shown. The longest semiaxis is four. The red input $v_2 + v_3$ has length $\sqrt{2}$, not one. Its output length $\sqrt{17}$ therefore does not qualify for the spectral-norm maximum; normalizing the input gives output length $\sqrt{17/2} < 4$.



## Spectral lower bound: rank k cannot preserve k+1 dimensions

Take any B with rank at most k, without assuming shared singular vectors. On $S = \operatorname{span}\{v_1, \ldots, v_{k+1}\}$, B has rank at most k. Rank–nullity gives a unit $z \in S$ with $Bz = 0$.

Write $z = \sum_{i=1}^{k+1} c_i v_i$, where $\sum_i c_i^2 = 1$. Orthogonality gives

$$\|(A - B)z\|^2 = \|Az\|^2 = \sum_{i=1}^{k+1} \sigma_i^2 c_i^2 \geq \sigma_{k+1}^2.$$

Thus $\|A - B\|_2 \geq \sigma_{k+1}$. In our example, even the nondiagonal $B = \begin{bmatrix} 9 & 9 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ admits $z = (1, -1, 0)^T/\sqrt{2}$. Then Bz=0 and $\|Az\|^2 = (81 + 16)/2 = 97/2 \geq 16$. The argument covers arbitrary candidates, not just diagonal ones.

## Frobenius lower bound: split each column at a right angle

Let P be the orthogonal projector onto $C(B)$. For each pair of columns $a_j, b_j$, $Pb_j = b_j$, so

$$a_j - b_j = (I - P)a_j + (Pa_j - b_j).$$

The first part is perpendicular to $C(B)$; the second lies in it. Apply Pythagoras column by column and sum:

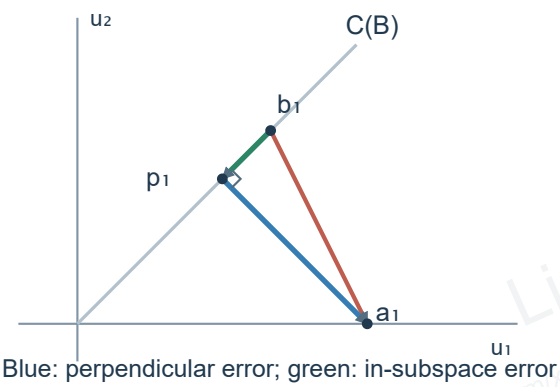
$$\|A - B\|_F^2 = \|(I - P)A\|_F^2 + \|PA - B\|_F^2 \geq \|(I - P)A\|_F^2.$$

Projection chooses the nearest point in the subspace. Choosing another point adds squared error. Complementary orthogonal projections also give

$$\|(I - P)A\|_F^2 = \|A\|_F^2 - \|PA\|_F^2 = \sum_{i=1}^{r} \sigma_i^2 - \operatorname{tr}(PAA^T).$$



## One projected column: 45 = 81/2 + 9/2




Blue: perpendicular error; green: in-subspace error

One column, in the first coordinate plane

$a_1$=(9,0,0), $b_1$=(6,6,0)

$p_1$=P$a_1$=(9/2,9/2,0)

$a_1 - b_1 = (a_1 - p_1) + (p_1 - b_1)$

$a_1 - p_1 \perp p_1 - b_1$

45 = 81/2 + 9/2

Use $C(B) = \operatorname{span}\{(1,1,0)^T\}$, $a_1 = (9,0,0)^T$, $b_1 = (6,6,0)^T$. Projection gives $p_1 = (9/2, 9/2, 0)^T$. Error $(3,-6,0)^T$ splits as $(9/2,-9/2,0)^T + (-3/2,-3/2,0)^T$. Their dot product is zero and squared lengths give $45 = 81/2 + 9/2$. This depicts one column; Frobenius error sums all columns.

## How much squared magnitude can a projection retain?

Complete left singular vectors to an orthonormal basis of $\mathbb{R}^m$, setting $\sigma_i = 0$ for i>r. Let $\alpha_i = u_i^T P u_i = \|Pu_i\|^2$. Then

$$0 \le \alpha_i \le 1, \qquad \sum_{i=1}^{m} \alpha_i = \operatorname{tr} P = \operatorname{rank} P \le k.$$

Think of $\alpha_i$ as projection capacity allocated to direction i: at most one each, at most k overall. Larger squared singular values contribute more per unit capacity. For k=1, the preceding projection gives:

| Direction | $\sigma_i^2$ | Capacity $\alpha_i$ | Retained squared magnitude |
|---|---|---|---|
| $u_1 = e_1$ | 81 | 1/2 | 81/2 |
| $u_2 = e_2$ | 16 | 1/2 | 8 |
| $u_3 = e_3$ | 1 | 0 | 0 |

Total capacity is one, yet retained magnitude is $97/2$, below 81 from allocating everything to the first direction. No other one-dimensional projection can exceed 81; next prove the general k bound.



## Prove the bound for retaining the largest k terms

Use $0 \le \alpha_i \le 1$, $\sum_i \alpha_i \le k$. Leading singular values are at least $\sigma_k$; all remaining ones are at most $\sigma_k$. Thus

$$\begin{aligned}\operatorname{tr}(PAA^T) &= \sum_{i=1}^{m} \sigma_i^2 \alpha_i \\ &\le \sum_{i=1}^{k} \sigma_i^2 \alpha_i + \sigma_k^2 \sum_{i=k+1}^{m} \alpha_i \\ &\le \sum_{i=1}^{k} \left[\sigma_i^2 \alpha_i + \sigma_k^2 (1 - \alpha_i)\right] \le \sum_{i=1}^{k} \sigma_i^2.\end{aligned}$$

Use tail values at most $\sigma_k$, total capacity at most k, and $\sigma_i^2 \ge \sigma_k^2$. Substitution yields $\|A - B\|_F^2 \ge \sum_{i=k+1}^{r} \sigma_i^2$.

## Eckart–Young: truncation attains both lower bounds

The preceding proofs cover every rank-at-most-k B. The truncated remainder E has exactly the discarded singular values, attaining both bounds:

$$\min_{\operatorname{rank} B \le k} \|A - B\|_2 = \sigma_{k+1}, \qquad \min_{\operatorname{rank} B \le k} \|A - B\|_F^2 = \sum_{i=k+1}^{r} \sigma_i^2.$$

| For $A = \operatorname{diag}(9, 4, 1)$ | Minimum spectral error | Minimum Frobenius error |
|---|---|---|
| k=0: only B=0 | 9 | $\sqrt{98}$ |
| k=1 | 4 | $\sqrt{17}$ |
| k=2 | 1 | 1 |
| k≥3: B=A is allowed | 0 | 0 |

The main proofs use $1 \le k < r$. At k=0 do not invoke $\sigma_k$: use the unique rank-zero matrix. At k≥r the error is zero. Appending a zero singular value leaves the earlier 9,4,1,0 example's k=1,2,3 errors unchanged.



## Uniqueness depends on which norm you minimize

For A=diag(9,4,1), take $B_c = \operatorname{diag}(c, 0, 0)$. For every $5 \le c \le 13$, Bc has rank one and

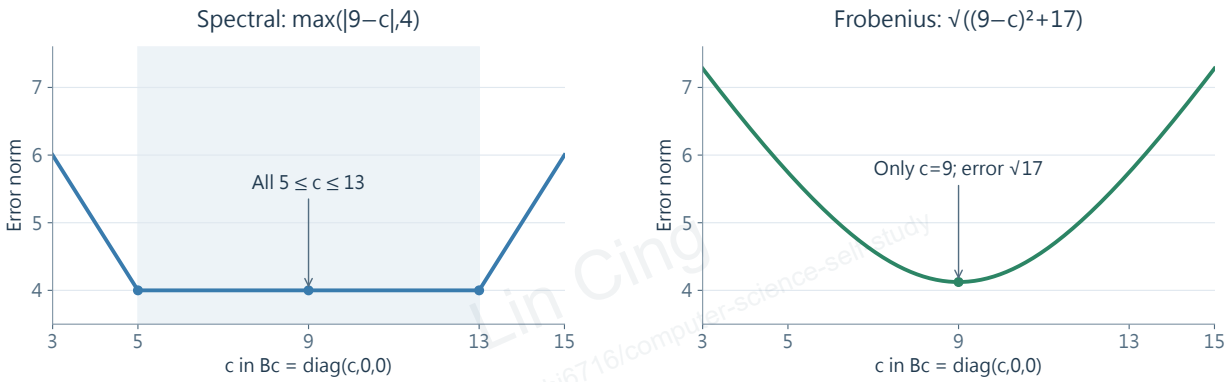
$$\|A - B_c\|_2 = \max\{|9 - c|, 4, 1\} = 4.$$

This whole family attains the global spectral lower bound despite the strict gap 9>4. In contrast,

$$\|A - B_c\|_F^2 = (9 - c)^2 + 17$$

is minimized within this family only at c=9. More generally, for $1 \le k < r$, a strict gap $\sigma_k > \sigma_{k+1}$ makes the Frobenius minimizer unique. Equality in the trace bound forces the first k capacities to one and all others to zero; equality in Pythagoras forces B=PA, hence B=$A_k$. If the cutoff splits a positive tie, different retained parts of that fixed complete singular subspace can give different optimum matrices.

## The same candidates, plotted under two objectives




Both plots use the same Bc family. On the left, the discarded second direction already causes error four; changing the first direction has no effect on the maximum while its error stays at most four. On the right, its squared error still counts, so only c=9 minimizes the total. This family illustrates nonuniqueness without claiming to list every spectral minimizer.



## Storage cost of low-rank compression

Rank-$k$ factors store

$$mk + nk + k = k(m + n + 1)$$

raw scalars, plus finite-precision metadata. Raw storage can decrease only if this count is below $mn$.

For a $1000 \times 1000$ image and rank 10, the factor count is $2.001 \times 10^4$, compared with $10^6$ entries. Actual formats also store headers, quantization information, and other metadata.

Random full-rank data often compress poorly. Compressibility is a property of the data, not a guarantee supplied by SVD.

## The data orientation in PCA

Let

$$A_0 \in \mathbb{R}^{m \times n},$$

with variables in rows and samples in columns. With row means $\mu_j$, center the data by

$$A_{js} = (A_0)_{js} - \mu_j.$$

The first principal direction $u_1$ maximizes the score vector's

$$s_1 = u_1^T A$$

sum of squares,

$$\|u_1^T A\|^2 = u_1^T A A^T u_1.$$

subject to $\|u_1\| = 1$. The spectral theorem gives maximum $\sigma_1^2$, attained at a dominant eigenvector of $AA^T$, the first left singular vector.



## Scores and sign ambiguity

From

$$A = U \Sigma V^T$$

the principal scores are

$$S = \Sigma V^T = U^T A.$$

Row $i$ lists all samples' scores on component $i$. Its squared norm equals $\sigma_i^2$, with zero rows included in a full representation.

Changing the signs of both $u_i$ and $v_i$ preserves the factorization. Changing only one changes the reconstruction. An individual sign has no intrinsic meaning.

Repeated singular values permit rotations of the component axes within the tied subspace. Compare subspace angles across PCA fits rather than matching arbitrary column numbers.

## Covariance denominators

For variables in rows and $n \geq 2$ samples, the usual sample covariance is

$$\widehat{C}_{n-1} = \frac{1}{n-1} AA^T,$$

The descriptive or population-denominator version is

$$\widehat{C}_n = \frac{1}{n} AA^T.$$

The eigenvectors coincide, while eigenvalues differ by factor $n/(n-1)$. For $n = 1$, the sample-variance denominator vanishes.

Unscaled PCA favors features with larger variance or larger numerical units. Standardization yields correlation-based PCA but changes the original geometry; it is a modeling choice.



## A two-dimensional PCA example

Samples:

$$(-1, -1), \quad (0, 1), \quad (1, 0).$$

Centered matrix:

$$A = \begin{bmatrix} -1 & 0 & 1 \\ -1 & 1 & 0 \end{bmatrix}.$$

$$AA^T = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}.$$

$$u_1 = (1, 1)^T/\sqrt{2}, \qquad u_2 = (1, -1)^T/\sqrt{2}.$$

Scores:

$$s_1 = \begin{bmatrix} -\sqrt{2} & 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix},$$
$$s_2 = \begin{bmatrix} 0 & -1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}.$$

The score rows are orthogonal and their squared norms are 3 and 1.

## How singular vectors determine ellipse axes

The input and output each use an orthonormal basis, generally with different directions. Take

$$V = \frac{1}{5}\begin{bmatrix} 4 & -3 \\ 3 & 4 \end{bmatrix}, \qquad U = \frac{1}{5}\begin{bmatrix} 3 & -4 \\ 4 & 3 \end{bmatrix}, \qquad \Sigma = \operatorname{diag}(3, 2).$$

Here $V^TV = U^TU = I$, and

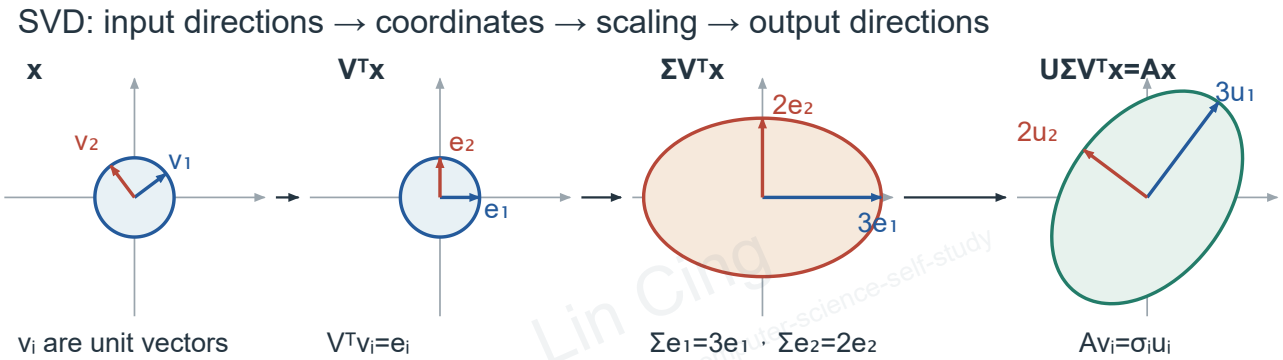
$$A = U\Sigma V^T = \frac{1}{5}\begin{bmatrix} 12 & -1 \\ 6 & 12 \end{bmatrix}.$$

All panels use the same coordinate scale. First $V^T$ maps input directions $v_i$ to $e_i$, then $\Sigma$ scales, and finally $U$ sets the output directions. Rotation preserves the circle as a set, but moves the marked vectors; that step must remain visible.

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## How singular vectors determine ellipse axes (continued)



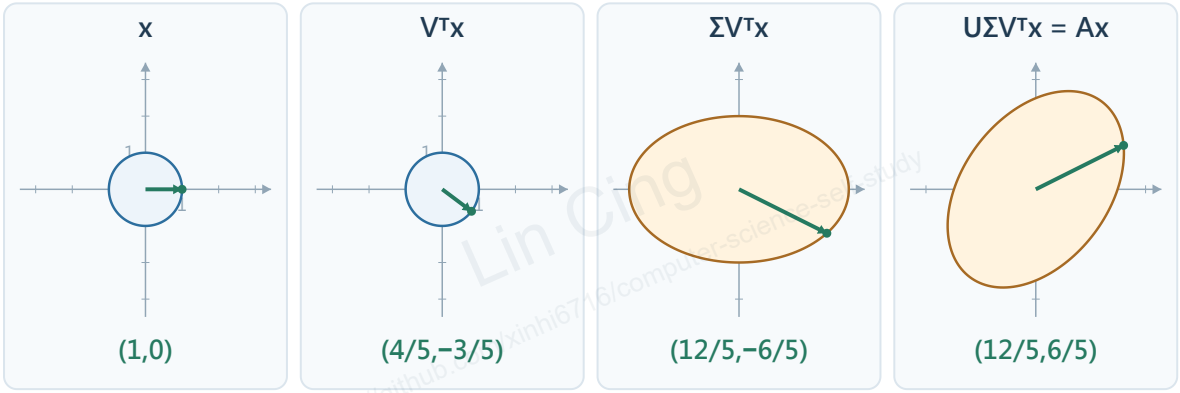

Check the two axes individually.

$$v_1 = (4/5, 3/5)^T, \quad Av_1 = (9/5, 12/5)^T = 3u_1,$$
$$v_2 = (-3/5, 4/5)^T, \quad Av_2 = (-8/5, 6/5)^T = 2u_2.$$

The semiaxis vectors are $3u_1, 2u_2$. The unit vectors $u_1, u_2$ specify directions. This full-rank example maps the unit circle to an ellipse boundary and the unit disk to the filled elliptical disk.

## Track a unit vector outside the singular directions

Keep the preceding U, V and Σ, and take $x = (1, 0)^T$, which is neither singular direction. The green arrow tracks this point; each background curve is the image of the unit circle at that stage.




Same scale in every panel; the green vector is the same input tracked through the map.



## Check coordinates and lengths separately

| Stage | Resulting coordinates | Squared norm |
|---|---|---|
| Input x | (1,0) | 1 |
| VᵀX | (4/5,−3/5) | 1 |
| ΣVᵀx | (12/5,−6/5) | 36/5 |
| UΣVᵀx=Ax | (12/5,6/5) | 36/5 |

Vᵀ and U preserve lengths; Σ performs the scaling. Here $\|Ax\| = 6/\sqrt{5}$, which is neither singular value.

For any unit input, write w=Vᵀx. Then $w_1^2 + w_2^2 = 1$, and

$$\|Ax\|^2 = 9w_1^2 + 4w_2^2 = 4 + 5w_1^2 \in [4, 9].$$

Thus $2 \le \|Ax\| \le 3$, with the upper bound at $x = \pm v_1$ and lower bound at $x = \pm v_2$. This argument covers every unit input; one numerical trace alone would not prove the bounds.

## SVD geometry with a right-angle rotation

For the unit ball $\|x\| \le 1$,

$$y = Ax = U\Sigma V^T x.$$

all three steps use the same $v_i, u_i$. Take

$$V = I, \quad \Sigma = \mathrm{diag}(3,2), \quad U = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}, \qquad A = U\Sigma V^T = \begin{bmatrix} 0 & -2 \\ 3 & 0 \end{bmatrix}.$$

Apply the factors in order.

1. $V^T x = Ix$ leaves input coordinates unchanged; the disk retains the coordinate-axis directions.
2. $\Sigma(V^T x) = (3x_1, 2x_2)$ scales the orthogonal axes, producing semiaxes 3 and 2.
3. $U\Sigma V^T x$ rotates the ellipse, sending the standard axes to $u_1 = (0,1)^T, u_2 = (-1,0)^T$.

The definitions give

$$v_1 = (1,0)^T, \quad Av_1 = (0,3)^T = 3u_1,$$
$$v_2 = (0,1)^T, \quad Av_2 = (-2,0)^T = 2u_2.$$



## SVD geometry with a right-angle rotation (continued)

Thus the ball maps to a filled ellipsoid in $C(A)$, with positive singular values as semiaxis lengths. In full-rank two dimensions, its boundary is an ellipse. With a nullspace, the unit sphere can map to the filled ellipsoid rather than only its surface. The maximum singular value is

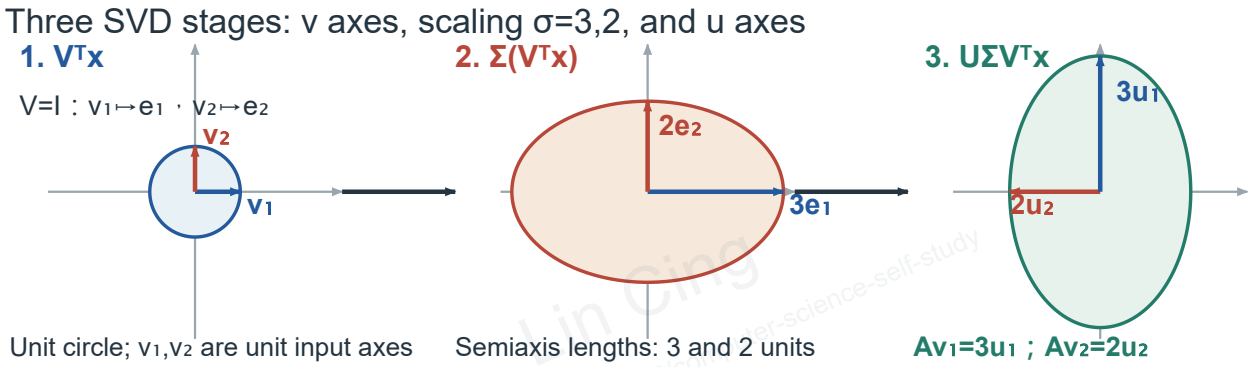
$$\|Av_1\| = \sigma_1.$$

Matrix norm：

$$\|A\|_2 = \sigma_1.$$

The smallest positive $\sigma_r$ controls the weakest growth from row space to column space.

## SVD geometry with a right-angle rotation (continued)




For deficient rank, take

$$B = \begin{bmatrix} 0 & 0 \\ 3 & 0 \end{bmatrix} = U \operatorname{diag}(3, 0) V^T.$$

Then $Bv_1 = (0, 3)^T$, while $Bv_2 = 0$. The $v_2$ direction collapses, and the unit disk maps onto the complete segment from $(0, -3)^T$ to $(0, 3)^T$.



## SVD geometry with a right-angle rotation (continued)

Rank-deficient SVD: B=[[0,0],[3,0]], σ2=0 collapses




**rank(B)=1; the full image is a line, the circle image a segment**

## Spectral norm and condition number

$$\|A\|_2 = \max_{x \neq 0} \frac{\|Ax\|_2}{\|x\|_2}.$$

Expand

$$x = \sum_i c_i v_i,$$

$$\|Ax\|^2 = \sum_i \sigma_i^2 |c_i|^2 \leq \sigma_1^2 \|x\|^2.$$

Equality holds at $x = v_1$, so

$$\|A\|_2 = \sigma_1.$$

For full column rank, the usual singular-value condition number is

$$\kappa_2(A) = \frac{\sigma_{\max}}{\sigma_{\min}}.$$

An orthogonal square matrix has every singular value equal to 1 and condition number 1. A rank-deficient matrix has a zero smallest singular value and no finite ordinary inverse condition number.



## Distance to singularity and numerical rank

For an invertible square $A \in \mathbb{R}^{n \times n}$, remove $\sigma_n u_n v_n^T$.

$$\|A - A_0\|_2 = \sigma_n.$$

Thus $\sigma_n$ is the spectral-norm distance to singularity. More generally, $A_k$ is a nearest rank-at-most-$k$ matrix, at distance $\sigma_{k+1}$.

Numerical rank uses a tolerance $\tau$.

$$\sigma_i > \tau \|A\|.$$

The tolerance must reflect noise, dimensions, and downstream sensitivity. Stable singular values do not imply stable individual singular vectors, especially near repeated or closely spaced values.

## Perturbation radius and numerical rank

For equally sized $A, B$, let $\delta = \|A - B\|_2$. Eckart–Young gives $\sigma_i(A) = \min_{\operatorname{rank} C < i} \|A - C\|_2$. Use the best low-rank approximation of $B$ and the triangle inequality.

$$\sigma_i(A) \le \|A - B\|_2 + \sigma_i(B).$$

Exchanging $A, B$ proves Weyl's bound $|\sigma_i(A) - \sigma_i(B)| \le \delta$. This is an absolute bound; relative errors in tiny singular values may still be large.

Suppose observed $A$ is within $\delta$ of an unknown true $B$, and let $r_\delta = \#\{i : \sigma_i(A) > \delta\}$. Every compatible $B$ has rank at least $r_\delta$, since those singular values stay positive. Conversely, $A_{r_\delta}$ has error $\sigma_{r_\delta+1} \le \delta$, so a compatible rank-$r_\delta$ model exists. If $r_\delta = q$, use $A_q = A$.

Thus $r_\delta$ is the minimum rank guaranteed within the perturbation ball, not the exact rank of every compatible model. For $A = \operatorname{diag}(1, 10^{-8}, 10^{-12})$, $\delta = 10^{-10}$, both a rank-2 truncation and the original rank-3 matrix are compatible. The data cannot identify whether the third direction is exactly zero.



## Perturbation radius and numerical rank (continued)

For a fixed absolute threshold $t$, classifications are robust only when $\sigma_i(A) > t + \delta$ or $\sigma_i(A) < t - \delta$. Directions near the threshold are uncertain. A relative threshold $t = \tau\|A\|_2$ also requires a recorded scale and noise model.

## QR pivots are not singular values

In exact arithmetic, $A = Q_1 R$, $Q_1^T Q_1 = I$, gives $A^T A = R^T R$, so $A$ and $R$ share singular values. However, the diagonal of $R$ is not its singular-value list. Take

$$R = \begin{bmatrix} 1 & K \\ 0 & 1 \end{bmatrix}, \qquad K \geq 1.$$

Both diagonal entries are 1, but $\|Re_2\|_2 = \sqrt{K^2 + 1}$, giving $\sigma_{\max} \geq K$. Since $\sigma_{\max}\sigma_{\min} = |\det R| = 1$, we obtain $\sigma_{\min} \leq 1/K$ and $\kappa_2(R) \geq K^2$. A large off-diagonal entry can conceal near dependence from the diagonal alone.

The sizes of $R_{ii}$ can be a heuristic, not a singular-value perturbation theorem. Column-pivoted QR often estimates numerical rank, but guaranteed accuracy requires explicit rank-revealing bounds or an SVD check. A nonzero floating-point pivot need not represent a trustworthy data direction.



## Polar decomposition

$$A = U\Sigma V^T = (UV^T)(V\Sigma V^T).$$

This page concerns square $A \in \mathbb{R}^{n \times n}$, so $U, V, \Sigma$ all have size $n \times n$. Set

$$Q = UV^T, \qquad S = V\Sigma V^T.$$

Then $Q$ is orthogonal and $S$ symmetric positive semidefinite, giving

$$A = QS.$$

If $A$ is invertible, every singular value is positive, so $S$ is positive definite. The sign of $\det Q$ records orientation reversal. The factors separate an orthogonal rotation or reflection from a symmetric stretch.

## The pseudoinverse

Reduced SVD:

$$A = U_r \Sigma_r V_r^T.$$

Moore-Penrose pseudoinverse:

$$A^+ = V_r \Sigma_r^{-1} U_r^T.$$

$$AA^+ = U_r U_r^T, \qquad A^+ A = V_r V_r^T.$$

These are the orthogonal projectors onto column space and row space, respectively.

Full column rank:

$$A^+ = (A^T A)^{-1} A^T.$$

Full row rank:

$$A^+ = A^T (AA^T)^{-1}.$$

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## The pseudoinverse (continued)

The vector $A^+ b$ is the minimum-norm least-squares solution. To prove this, write $x = \sum_i c_i v_i$ in the full right singular basis. Then $\|b - Ax\|^2 = \sum_{i=1}^{r} (u_i^T b - \sigma_i c_i)^2 + \sum_{i=r+1}^{m} (u_i^T b)^2$. Minimization requires $c_i = u_i^T b / \sigma_i$ for $i \leq r$. Nullspace coefficients remain arbitrary. Setting them to zero uniquely minimizes $\|x\|^2 = \sum_i c_i^2$, giving $A^+ b$.

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## Rank-deficient least squares

$$A = \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix} = (1,2)^T (1,1).$$

Normalize:

$$u_1 = (1,2)^T/\sqrt{5}, \qquad v_1 = (1,1)^T/\sqrt{2}.$$
$$A = \sqrt{10}\, u_1 v_1^T.$$

Here $u_1$ is an output direction and $v_1$ an input direction. Both have two coordinates in this example but have different values. For a rectangular matrix even their dimensions can differ. Rank one does not imply $u_1 = v_1$.

For $b = (1,3)^T$,

$$A^T A = \begin{bmatrix} 5 & 5 \\ 5 & 5 \end{bmatrix}, \qquad A^T b = \begin{bmatrix} 7 \\ 7 \end{bmatrix}.$$

Every parameter satisfying

$$x_1 + x_2 = 7/5$$



## Rank-deficient least squares (continued)

is a minimizer. The normal matrix is singular and cannot be inverted.

## The minimum-norm solution

$$\sigma_1 u_1^T b = \sqrt{10}\frac{1+6}{\sqrt{5}} = 7\sqrt{2}.$$

$$x^+ = A^+ b = \frac{1}{\sigma_1} v_1 u_1^T b = \frac{7}{10}(1,1)^T.$$

Projection:

$$AA^+ b = u_1 u_1^T b = \frac{7}{5}(1,2)^T.$$

Residual:

$$e = b - AA^+ b = \left(-\frac{2}{5}, \frac{1}{5}\right)^T.$$

$$e^T(1,2) = 0,$$

The residual is therefore orthogonal to column space.



## Numerical workflow

SVD workflow:

1. Check shape, missing values, and units.
2. Center PCA data and standardize when the modeling choice requires it.
3. Use a stable SVD algorithm without forming normal equations.
4. Inspect singular-value gaps and numerical rank.
5. Choose $k$ from the error budget.
6. Validate on held-out data or the downstream task.

Signs can flip and tied axes can rotate. Linear reconstruction alone establishes neither causal importance nor fairness.

## The four subspaces

| Space | Ambient space | SVD basis | Dimension |
|---|---|---|---|
| row space | $\mathbb{R}^n$ | $v_1, \dots, v_r$ | $r$ |
| nullspace | $\mathbb{R}^n$ | $v_{r+1}, \dots, v_n$ | $n - r$ |
| column space | $\mathbb{R}^m$ | $u_1, \dots, u_r$ | $r$ |
| left nullspace | $\mathbb{R}^m$ | $u_{r+1}, \dots, u_m$ | $m - r$ |

Input and output singular vectors belong to different roles and cannot generally be paired by an inner product. Equal dimensions allow an explicit coordinate identification, but do not make the bases identical. SVD pairs them through

$$Av_i = \sigma_i u_i.$$



## Chapter check

You can now do the following.

- Construct $v_i$ from $A^TA$ and set $u_i = Av_i/\sigma_i$.
- Complete the null directions to obtain a full SVD.
- Compute spectral and Frobenius truncation errors.
- Explain PCA centering, scores, variance, and denominator choices.
- Distinguish input and output singular bases.
- Derive the operator norm, condition number, and distance to singularity.
- Construct polar factors and a pseudoinverse.
- Handle sign ambiguity, ties, and numerical rank.

Chapter 8 reinterprets transformations through bases and coordinates.

## References and attribution

Gilbert Strang. Introduction to Linear Algebra, 5th Edition, 2016.

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